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%%writefile vector_add.cu
#include <iostream>
#include <cuda.h>

using namespace std;

// CUDA Kernel
__global__ void vectorAdd(float *A, float *B, float *C, int n) {
    int i = blockIdx.x * blockDim.x + threadIdx.x;
    if (i < n) {
        C[i] = A[i] + B[i];
    }
}

int main() {
    // Hardcoded for 1 million elements
    int n = 1000000;
    size_t size = n * sizeof(float);

    // Allocate memory on the Host (CPU)
    float *h_A = (float *)malloc(size);
    float *h_B = (float *)malloc(size);
    float *h_C = (float *)malloc(size);

    // Initialize vectors with hardcoded data
    for (int i = 0; i < n; i++) {
        h_A[i] = 1.0f;
        h_B[i] = 2.0f;
    }

    // Allocate memory on the Device (GPU)
    float *d_A, *d_B, *d_C;
    cudaMalloc((void **)&d_A, size);
    cudaMalloc((void **)&d_B, size);
    cudaMalloc((void **)&d_C, size);

    // Copy data to GPU
    cudaMemcpy(d_A, h_A, size, cudaMemcpyHostToDevice);
    cudaMemcpy(d_B, h_B, size, cudaMemcpyHostToDevice);

    int threadsPerBlock = 256;
    int blocksPerGrid = (n + threadsPerBlock - 1) / threadsPerBlock;

    // Timer
    cudaEvent_t start, stop;
    cudaEventCreate(&start);
    cudaEventCreate(&stop);

    // Launch Kernel
    cudaEventRecord(start);

```

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    cudaEventRecord(start),
    vectorAdd<<<blocksPerGrid, threadsPerBlock>>>>(d_A, d_B, d_C, n);
    cudaEventRecord(stop);
    cudaEventSynchronize(stop);

    float ms = 0;
    cudaEventElapsedTime(&ms, start, stop);

    // Copy result back to CPU
    cudaMemcpy(h_C, d_C, size, cudaMemcpyDeviceToHost);

    // Verify first 5 results
    cout << "--- First 5 Results ---\n";
    for (int i = 0; i < 5; i++) {
        cout << h_A[i] << " + " << h_B[i] << " = " << h_C[i] << "\n"
    }
    cout << "... \n[Success] Processed " << n << " elements.\n";
    cout << "[TIME] GPU Execution Time: " << ms << " ms\n";

    cudaFree(d_A); cudaFree(d_B); cudaFree(d_C);
    free(h_A); free(h_B); free(h_C);

    return 0;
}

```

Writing vector_add.cu

```

!nvcc vector_add.cu -o vector_add
!./vector_add

```

```

nvcc warning : Support for offline compilation for architectures prior to '<comp
--- First 5 Results ---
1 + 2 = 3
1 + 2 = 3
1 + 2 = 3
1 + 2 = 3
1 + 2 = 3
...
[Success] Processed 1000000 elements.
[TIME] GPU Execution Time: 99.9829 ms

```

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